SG No. 146 C_3^4 $R3$ [trigonal]

* generator : $\{3^+_{001}|0\}$
* symmetry operation $+[0,\,0,\,0]$, $+\left[\frac{2}{3},\,\frac{1}{3},\,\frac{1}{3}\right]$, $+\left[\frac{1}{3},\,\frac{2}{3},\,\frac{2}{3}\right]$

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det
1	$\{1\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1
2	$\{3^+_{001}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1
3	$\{3^-_{001}\|0\}$	$\begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1